

Backward Kernels for Block-Controlled Tail Viability

Exact Young-Envelope Preimages, Reset Criteria, and Two Unavoidable Birth Walls

Prime Dynamics Theory Program

July 2026

Abstract

RH-139 reduced eventual directional support to a controlled tail tube $\limsup y_n < 1$ and a positive normalized-base $\liminf$. RH-137 propagated a greedy finite trajectory forward. We develop the dual backward theory. For

$$F_{A,B,q}(y) = q + (\sqrt{Ay} + \sqrt{B})^2$$

and target radius r , the largest input radius mapped strictly below r is

$$\Phi_{A,B,q}(r) = \begin{cases} 0, & r \leq q + B, \\ 1, & A = 0 \text{ and } r > q + B, \\ \min \left\{ 1, \frac{(\sqrt{r-q} - \sqrt{B})^2}{A} \right\}, & A > 0, r > q + B. \end{cases}$$

For a finite control family, the predecessor radius is the maximum of these quantities. Backward composition gives the exact scalar viability kernel and a constructive policy. At block boundaries, if the backward kernel of every block contains a common interval $[0, r)$, then that interval is controlled invariant across arbitrarily many blocks; all intermediate safety constraints are already included.

Recomputing the full RH-137 candidate families gives 28 of 30 chains with initial kernel $[0, 1)$ and safe policies even from 0.99. The two coarse left failures have empty initial kernels. At their decisive step, every candidate has $F(0) > 9.4 > 1$, so no gauge choice or incoming state can cross the wall. This strengthens the prior classification: those events are candidate-family obstructions, not greedy artifacts. It also isolates them as finite-prefix births; no repeating all-level block condition is proved.

1 Forward envelopes and backward questions

At time n , RH-137 provides a finite family of monotone Young envelopes

$$F_{n,c}(y) = q_{n,c} + (\sqrt{A_{n,c}y} + \sqrt{B_{n,c}})^2, \quad c \in \mathcal{C}_n.$$

A forward policy chooses c_n and propagates an upper state. This answers whether one chosen trajectory is safe, but it does not directly answer two structural questions: what is the full set

of states from which some future policy remains safe, and is a failed greedy step unavoidable for the entire control family?

Because every envelope is scalar and increasing, both questions have exact answers by backward dynamic programming [Aubin, 1991, Bertsekas, 2017]. Unlike a per-step contraction test, the backward kernel automatically allows an expanding step when later constraints still leave enough room.

2 Exact one-step preimages

Fix the safety domain $0 \leq y < 1$, although the formulas hold for any target radius.

Theorem 1 (Young-envelope preimage). *For $A, B, q \geq 0$ and target $r \geq 0$, the supremum of*

$$\{y \in [0, 1] : F_{A,B,q}(y) < r\}$$

is $\Phi_{A,B,q}(r)$ displayed in the abstract. In particular the preimage is empty exactly when the zero-state floor $q + B$ is at least r .

Proof. The map is increasing. If $r \leq q + B = F(0)$, no input works. If $A = 0$, the map is constant and every input works once its floor is below r . Otherwise the inequality is equivalent to

$$\sqrt{Ay} < \sqrt{r - q} - \sqrt{B}.$$

The right side is positive precisely in the remaining case. Squaring and capping at one gives the formula. $\square$

For a control family define the predecessor operator

$$\mathcal{V}_n(r) = \max_{c \in \mathcal{C}_n} \Phi_{A_{n,c}, B_{n,c}, q_{n,c}}(r).$$

The maximizing control is a constructive witness.

Theorem 2 (Finite controlled kernel). *For a horizon N with safety wall one, set $r_N = 1$ and*

$$r_n = \mathcal{V}_n(r_{n+1}), \quad n = N - 1, \dots, 0.$$

Then $[0, r_n)$ is exactly the set of scalar upper states at time n from which some policy keeps every later state below one through time N .

Proof. At the last step the claim is Theorem 1, after maximizing over controls. Inductively, a state is viable at time n exactly when one control maps it into the already characterized interval $[0, r_{n+1})$. Monotonicity makes the union of all controlled preimages the interval ending at their largest radius. Backward induction proves exactness. $\square$

classification	count	strongest finite statement
viable	28	full initial interval $[0, 1)$; near-boundary policy safe
obstructed	2	empty kernel; every control floor exceeds target
all chains	30	backward and forward classifications agree

Table 1: Exact scalar backward-kernel classification of the RH-137 families.

3 Blocks, resets, and delayed starts

Partition an infinite recurrence into blocks $[n_j, n_{j+1})$. Let $\mathcal{V}^{(j)}$ denote the backward operator for the whole block, including every intermediate safety wall.

Theorem 3 (Repeating-block invariance). *Suppose there is $r > 0$ such that, for every j beyond some index,*

$$\mathcal{V}^{(j)}(r) \geq r.$$

Then every boundary state in $[0, r)$ admits a control policy that remains below one throughout each block and returns to $[0, r)$ at the next boundary. Consequently the controlled trajectory has $\limsup y_n < 1$ if all partial block images of $[0, r)$ are uniformly bounded by some $\bar{y} < 1$.

Proof. The inclusion $[0, r) \subseteq [0, \mathcal{V}^{(j)}(r))$ says exactly that every boundary state has a within-block policy ending below r while satisfying all intermediate walls. Concatenate these policies. The uniform partial guard gives the stated limsup. $\square$

A reset that places the state below r may start this induction after an arbitrary finite prefix. Conversely, an endpoint condition alone is not enough: a two-step block can jump above one and later return to zero. The backward block operator avoids this error because every intermediate target is imposed during composition.

4 Finite candidate-family audit

The audit reconstructs every RH-137 temporal record and every polar/blended gauge candidate at 80 decimal digits. It then applies the exact scalar preimage formula backward, stores the maximizing controls, and verifies the result by starting at $0.99r_0$ and propagating the selected policy forward.

Among the viable chains, the narrowest positive interior reset radius is 1.724×10^{-3} . Thus a chain can accept the whole initial interval because early zero-tail steps act as resets, yet still pass through a much narrower later gate. This distinction is precisely what block analysis is meant to record.

The two empty kernels are the left channel at $\sigma = 0.08$ for thresholds 10^{-8} and 10^{-6} . At the decisive final transition the birth term alone exceeds 9.4. Since

$$\min_{c \in \mathcal{C}_n} F_{n,c}(0) > 9.4 > 1,$$

the predecessor of the unit safety interval is empty. No incoming tail, orthogonal gauge, Young parameter, or non-greedy future choice can repair that step within this candidate architecture.

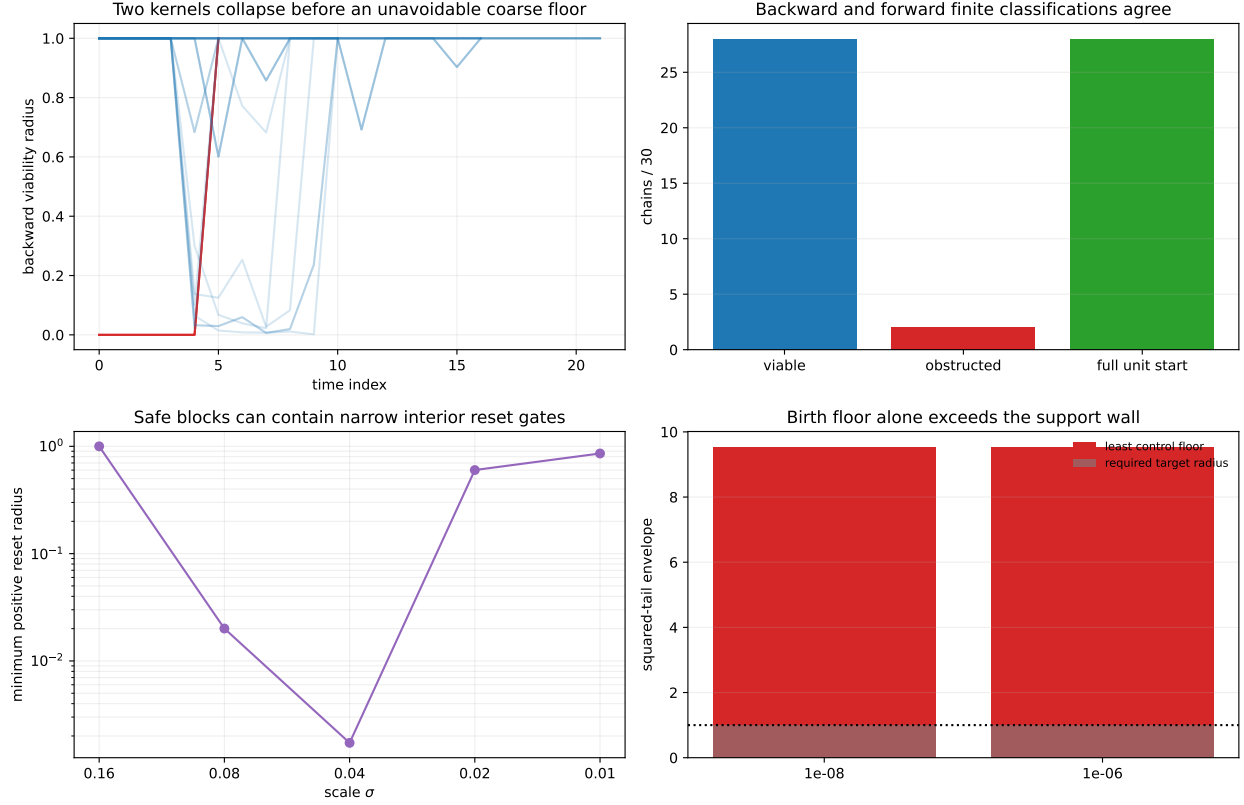


Figure 1: Backward kernels, finite classification, interior reset radii, and the two unavoidable zero-state floors.

5 Consequence and claim boundary

RH-144 gives an exact method for proving the RH-139 tail packet P_V from repeating controlled blocks. It also shows that the two finite failures are localized more strongly than before: they are control-family walls, not poor greedy choices. Since they occur at one coarse anchor, they do not by themselves preclude a delayed-start asymptotic theorem.

We have not proved that all finer or all future model blocks share a common positive backward radius, established a uniform partial-block gap below one, or justified a delayed start for every all-level source realization. We have not proved the normalized-base $\liminf$, Stage A, a Hilbert–Polya operator, a zeta-zero identification, or the Riemann Hypothesis.

References

- Jean-Pierre Aubin. *Viability Theory*. Birkhäuser, 1991.
- Dimitri P. Bertsekas. *Dynamic Programming and Optimal Control*. Athena Scientific, 2017.